A structured synthetic dataset of English and Italian verb alternations for testing lexical abstraction via functional lexicon in LLMs.

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Abstract

In this paper, we develop a structured dataset the investigation of lexical abstraction via functional words in LLMs, concentrating on the change-of-state (COS) and object-drop (OD) verb alternations classes in English and Italian. We present the dataset creation process and reports the results of baseline experiments.

1 Introduction

Artificial Neural Networks (ANNs) have demonstrated success in many language-based applications, and the nature of their linguistic knowledge have been extensively investigated (Warstadt et al., 2019; Ravichander et al., 2021; Belinkov, 2022; Acs et al., 2024). These models typically operate on raw textual input, processing language as sequences of discrete word or subword units, which constitute the fundamental building blocks for computational approaches to natural language (Stevenson and Merlo, 2022). However, not all words in grammars function in the very same way, and natural languages' lexicons contain elements with special properties, such as pronominal forms such as English *they*, *this*, *somebody* or their Italian equivalent *loro*, *questo* e *qualcuno*. These words are semantically deficient in terms of denotation when used out of context, despite possessing the relevant properties to convey described event meanings. Following previous literature (Chomsky, 1995; Hudson, 1997), we refer to these as *functional words* (henceforth, FUN).

Formal theories, as well traditional grammars (Albrow, 1972), contrast this class of elements to more referential, contentive words (Cann, 1999) such as nominal entities (e.g. English *the professor*, *the books*, *bread* and their Italian equivalent *la professoressa*, *i libri* and *pane*) which we refer to as lexical elements (LEX). Works in psycholinguistics have shown differences in parsing between these two classes across linguistic phenomena and structures in adult grammars (Bever, 1974; Belletti and Rizzi, 2013; Chesi and Canal, 2019), developmental grammar (Friedmann et al., 2009) and language pathologies (Mantione, 2016). Studies in formal semantics have sought to understand the relationship between these two classes, typically interpreting pronouns (Pron) in two primary ways.

Pronouns can be treated either (i) as variables—placeholders for more structured lexical elements within a sentence—or (ii) as equivalent to noun phrase (NP) descriptions, referring to fuller expressions in context (cf. Büring 2019).[1]

The relationship between these two entities becomes of interest at the interface of syntax and semantics, particularly when analyzing complex verbal configurations where the argument structure encodes semantic elements of the arguments (Dowty, 1989; Rappaport and Levin, 1988; Levin, 1993). We illustrate this phenomenon using the English example in Figure 1, featuring alternating verbs (i.e., verb forms that can occur in both transitive and intransitive forms) such as *paint* and *break*. These verbs differ in the argument structure of their intransitive forms. In pragmatically appropriate contexts, the NPs *the artist* and *this vase* can be replaced by corresponding pronominal and deictic forms (e.g., *she* or *this*), where the choice of pronoun aligns with the argument structure properties of the verb class (an agent or a patient).

Structure	Verb *paint*	Verb *break*
Transitive Lex	*The artist was painting this vase*	*The artist broke this vase*
Intransitive Fun	*She was painting*	*This* broke/*She broke

Figure 1: Example of Lexical (Lex) and Functional (Fun) elements involving the English verb *break* and the English verb *paint* in transitive and intransitive configurations.

Other languages, such as Italian, morphologically mark the intransitive form of verbs like rompere (the equivalent of English *break*), typically using a reflexive-like element (*si*, in bold in Figure 2) when the patient of the transitive verb becomes the subject of the intransitive. Italian also allows for a special form of pronominalization in these contexts—specifically, null subjects (pro), as discussed in Rizzi (1982, Ch. 4), represented in brackets in Figure 2.

Structure	Verb *paint*	Verb *break*
Transitive Lex	*L'artista stava dipigendo questo vaso*	*L'artista ruppe queste vaso*
Intransitive Fun	*(Lei) stava dipingendo*	*Questo* **si** ruppe / *(lei) ruppe

Figure 2: Example of Lexical (Lex) and Functional (Fun) elements involving the Italian verb *rompere* 'break' and the verb *dipingere* 'paint' in transitive and intransitive configurations.

The verbs adopted as exemplars in Figures 1 and 2 belong to two distinct, well-studied verb classes (Levin, 1993): (i) Change-of-State (COS) verbs (e.g., *break*), and (ii) Object-Drop (OD) verbs (e.g., *paint*) (Merlo and Stevenson, 2001). These two semantic classes of verbs differ in their idiosyncratic lexical meaning, but only minimally in their argument syntactic structure properties. In COS the subject of the intransitive is the patient of the transitive, while in OD the subject of the intransitive is the subject of the transitive.

[1] In this article, we do not commit to either theoretical approach.

To investigate whether language models possess knowledge about verbal argument structure, numerous studies have employed transformer models (Kann et al., 2019; Thrush et al., 2020; Yi et al., 2022; Wilson et al., 2023). While most of these studies have focused on comparing metrics derived from structural minimal pairs, Samo et al. (2023) developed a more complex experimental setup to determine whether argument structure alternations are learned—specifically, the English spray-load alternation. Instead of relying on minimal sentence comparisons, they constructed a structured, curated dataset using Blackbird Language Matrices (BLM; Merlo (2023)), a novel linguistic task inspired by Raven Progressive Matrices Raven (1938) (RPMs). BLM provides a framework for evaluating the learning of linguistic paradigms in ANNs by requiring models to identify relevant objects and attributes and organize them into a logical progression to arrive at a predictive solution. In their dataset, Samo et al. (2023) decomposed the alternation into its constituent components to provide the model with all necessary information to solve the task. Building on this methodology, we present the construction of a new dataset and baseline experiments to systematically compare Change-of-State (COS) and Object-Drop (OD) alternations between English and Italian. The steps are presented in section 2. We also run some baseline experiments, presented in seection 3.

2 Data

We developed manually curated pairwise contrastive datasets with respect to the variables (Lex vs. Fun, OD vs. COS) in both English and Italian. Each of the datasets comes in three variants of different amounts of lexicalisation in the spirit of previous works on BLM (An et al., 2023; Samo et al., 2023). The BLM task is a sequence-completion, multiple-choice task where, given a context, a correct answer must be chosen among others to continue the sequence in the context. The context sequence comprises sentences that are examples of the same linguistic phenomenon or grammatical rule and covertly provides all the relevant information for continuation. The answer set consists of minimally differing sentences from which the unique continuation is to be chosen. The contrastive alternatives illustrate either mistakes on the underlying linguistic phenomenon or mistakes concerning the sequence.

Because of its double-structure nature, a BLM is more informative for our investigation of abstraction than simply studying individual sentences. The BLM allows us to study the abstraction in the sentences that compose it but also if and how this abstraction is used in solving the BLM task.

Verb alternations are linguistic phenomena that lend themselves well to the construction of a BLM because the alternations are systematic and observable (one or two arguments in different syntactic positions), but correspond to unobservable attributes (agent, patient), and their combinatorial distribution. Additionally, the fact that these constructions do not affect any adjunct elements allows for the insertion of linguistic elements unrelated to the phenomenon, which is an important factor for the construction of BLM templates

(Merlo, 2023). These phenomenon-external elements serve as a testbed for the underlying logical reasoning part of the BLM task.

Among verb alternations, causatives and object-drop are interesting because they provide an argument structure minimal pair: they share the same syntactic structure but differ in their underlying argument structure.

2.1 BLM template

Context The syntax-semantics features of the verb alternations, and their combination rules, lead to the construction of the context set.

Specifically, (i) the presence of one or two arguments and their attributes (agents, Ag; patients, Pat) ; (ii) the active (Akt) or passive (Pass) voice of the verb. The phenomenon-external factors related to logical reasoning involve the number and quality of nominal phrases (NP) following the verb and its core arguments. This includes an alternation every two context sentences between a NP introduced (i) by any preposition (e.g., *in an instant*, henceforth p-NP) and (ii) by the preposition by (e.g., *by chance*, by-NP), but not agentive (e.g., *by the artist*, by-Ag/by-Pat), which remains a confounding variable.

The rationale behind this specific, complex BLM structure under investigation is that the progression is designed for the COS, rather than the OD. In particular, the passive voice serves as an intermediate stage for the intransitive (where the patient is the subject, though marked with passive voice), while it acts as a confounding element for the OD.[2]

Due to the asymmetry between the two classes of verbs, the COS-BLM and OD-BLM minimally differ with respect to an intransitive sentence in the context, the intransitive one followed by p-NP. The minimal difference between the two classes is the semantic attribute of the subject of the intransitive representing the last sentence of the context (sentence 7 in the contexts in Figure 3). Naturally, also the correct answer varies across the two groups, but they still structurally represent an intransitive form followed by by-NP.

Answer Set The solution to the BLM task is a sentence that follows both the rules of the linguistic phenomenon under investigation and at least one phenomenon-external rule.

In our case, the rule is the understanding of the missing pair with respect to the form of the verb, transitive and intransitive verbs, interspersed with passive structures and the unrelated (phenomenon-external) rule is the alternation between p-NP and by-NP as last constituent. The answer set is therefore built on the full understanding or lack of understanding of all the underlying rules. Consequently, the answer set is composed of a correct answer and a series of wrong answers, which all deviate from the target answer, containing a qualitative different set of mistakes. All the answers have the same structure, consisting of a verb, two nominal constituents (giving rise to a structure of the type NP V NP)

[2]The idea of missing relevant progression to irrelevant progression is borrowed from studies in RPM and how they affect learning (Meo et al., 2007).

and a preposition (by, or the lack of the preposition) between the verb and the second NP, for a total of four linguistic elements (NP V by-NP).

The structure of the answer set is the place where we can fundamentally understand whether the model has learnt linguistic information. Each answer represents a particular type of violation of the rules underlying the construction of the BLM structure. The error labeled SSM-Int represents an intransitive structure, differing from the target answers because an agent is the subject in a COS intransitive construction, while a patient is the subject in an OD intransitive construction. This error follows an imperfect syntax-semantics mapping: building on Samo et al. (2023), we adopt the label SSM for this kind of error. In Passive answers (Pass), the verb is not inflected for the correct voice (an active voice is expected), resulting in a passive structure. Transitive structure (Trans) are violations of the logical reasoning of the BLM, and they result in a transitive sentence without a PP. Finally, WrBy are cases of ungrammatical sentences which only mininally differ from the target answer by the lexical choice of NP (the other argument) which follows the preposition *by*. Pass, Trans and WrBy may also contain errors in syntax-semantic mapping (when the subject of the answer is the wrong argument) and therefore their label is preceded by SSM (e.g. SSM-Pass, SSM-trans, SSM-WrBy). The templates for the context and the answer set for both sets of datasets are provided in Table 3.

Italian data provides two additional answers, which are related to the presence or absence of the fake reflexive *si*, typical of causative constructions but not of OD. An important difference is that, while the lack of *si* results in ungrammatical clauses for COS (e.g., *il burro si scioglie* 'the butter melts' vs. **il burro scioglie*), the presence of *si* in OD configurations is interpreted by speakers as a pure reflexive (*le artiste dipingevano* 'the artists were painting' vs. *le artiste si dipingevano* 'the artists were painting themselves'). Therefore, the sentence forms a grammatical structure, but not an intransitive clause — it is instead a transitive one.

Note that the answer set does not change across verb classes. For example, the correct answer for COS is an error for OD, while the correct answer for OD is considered a mistake for COS. Italian data offer a wider selection of answers to detect changes with the morpho-syntactic elements.

2.2 FUN vs. LEX

An abstract component in natural languages is the functional lexicon (FUN), such as pronominal forms (e.g., *I, you, her*) and demonstratives (*this, that*). Less abstract linguistic elements will include lexical items of varying complexity for agents, patients, and the other nominal phrases introduced by prepositions (LEX). We illustrate the two typologies in Figure 4 with the English verb *break*.

In line with previous works on BLMs, each dataset also contains a varying amount of lexicalisation to increase complexity progressively. We add, respectively, the label of type I, type II and type III. In type I, the lexical material of the sentences do not change across the context and the answer set; in type II,

COS context				
1	Ag	Akt	Pat	p-NP
2	Ag	Akt	Pat	by-NP
3	Pat	Pass	by-Ag	p-NP
4	Pat	Pass	by-Ag	by-NP
5	Pat	Pass		p-NP
6	Pat	Pass		by-NP
7	Pat	Akt		p-NP
?	???			

COS answers		
1	Pat Akt by-NP	**Correct**
2	Ag Akt by-NP	SSM-Int
3	Pat Pass by-Ag	ER-Pass
4	Ag Pass by-Pat	SSM-Pass
5	Pat Akt Ag	R-Trans
6	Ag Akt Pat	SSM-Trans
7	Pat Akt by-Ag	E-WrBy
8	Ag Akt by-Pat	SSM-WrBy

OD context				
1	Ag	Akt	Pat	p-NP
2	Ag	Akt	Pat	by-NP
3	Pat	Pass	by-Ag	p-NP
4	Pat	Pass	by-Ag	by-NP
5	Pat	Pass		p-NP
6	Pat	Pass		by-NP
7	Ag	Akt		p-NP
?	???			

OD answers		
1	Pat Akt by-NP	SSM-Int
2	Ag Akt by-NP	**Correct**
3	Pat Pass by-Ag	SSM-Pass
4	Ag Pass by-Pat	Pass
5	Pat Akt Ag	SSM-Trans
6	Ag Akt Pat	Trans
7	Pat Akt by-Ag	SSM-WrBy
8	Ag Akt by-Pat	WrBy

Figure 3: BLM context sets (left) for the change-of-state group (COS) and the object drop (OD) class and answers sets (right). Italian COS and OD has two additional answers in which the presence/absence of the reflexive-like element *si* is absent (NoSI/SSM-NoSI in COS) or present (SI/SSM-SI in OD).

the verb remains the same while the constituents vary across the context and the answer set; finally, in type III data, we have extreme lexical variation in the context sentences and the answer set. The types are exemplified in the relevant section of Figure 5, together with the generation process presented in the next paragraph.

Process of Generation We selected 30 verbs from each of the two classes discussed in Levin (1993). These verbs were manually chosen from English.[3] The choice depended on whether their Italian translations maintained the same structure.[4]

The functional lexicon for both English and Italian have been manually selected by the author to maintain acceptability of the sentences both syntactically and semantically[5]. The semi-automatic generation of the synthetic lexical data started from the English data: they were manually selected from a list of the

[3]English COS: *bake*, *bend*, *blacken*, *break*, *brighten*, *caramelize*, *chip*, *close*, *corrode*, *crinkle*, *defrost*, *empty*, *expand*, *fry*, *harden*, *harmonize*, *heat*, *improve*, *increase*, *intensify*, *melt*, *open*, *propagate*, *purify*, *sharpen*, *shrink*, *sweeten*, *tear*, *whiten*, *widen*; OD: *clean*, *cook*, *draw*, *drink*, *eat*, *fish*, *hum*, *iron*, *knead*, *knit*, *mend*, *milk*, *nurse*, *paint*, *play*, *plow*, *polish*, *read*, *recite*, *sculpt*, *sew*, *sing*, *sow*, *study*, *sweep*, *teach*, *wash*, *weave*, *whittle*, *write*.

[4]Italian COS: *addolcire*, *affilare*, *allargare*, *annerire*, *aprire*, *armonizzare*, *caramellare*, *chiudere*, *corrodere*, *cuocere*, *espandere*, *friggere*, *illuminare*, *indurire*, *ingrandire*, *intensificare*, *migliorare*, *piegare*, *propagare*, *purificare*, *riscaldare*, *rimpicciolire*, *rompere*, *scheggiare*, *sciogliere*, *scongelare*, *sbiancare*, *stropicciare*, *svuotare*; OD: *allattare*, *arare*, *bere*, *cantare*, *canticchiare*, *cucire*, *cucinare*, *dipingere*, *disegnare*, *impastare*, *insegnare*, *intagliare*, *lavare*, *leggere*, *lucidare*, *mangiare*, *mungere*, *pescare*, *giocare*, *rammendare*, *recitare*, *saldare*, *scolpire*, *seminare*, *spazzare*, *stirare*, *studiare*, *tessere*, *scrivere*.

[5]Following the discussion in Haspelmath (1997), we add elements like *somebody* as pronominal elements.

	EnCOSLex
1	The witch can break an oath within seconds
2	The witch can break an oath by chance
3	An oath can be broken by the witch within seconds
4	An oath can be broken by the witch by chance
5	An oath can be broken within seconds
6	An oath can be broken by chance
7	An oath can break within seconds
?	???

	EnCOSLex
1	**An oath can break by chance**
2	The witch can break by chance
3	An oath can be broken by the witch
4	The witch can be broken by an oath
5	An oath can break the witch
6	The witch can break an oath
7	An oath can break by the witch
8	The witch can break by an oath

	EnCOSLex
1	She can break it with this
2	She can break it by that
3	It can be broken by her with this
4	It can be broken by her by that
5	It can be broken with this
6	It can be broken by that
7	It can break with this
?	???

	EnCOSLex
1	**It can break by that**
2	She can break by that
3	It can be broken by her
4	She can be broken by it
5	It can break her
6	She can break it
7	It can break by her
8	She can break by it

Figure 4: Examples of Fun and Lex for the English verb *break*, a COS verb

top twenty-five results automatically generated using masked modeling. The language model is different (*bert-base-uncased*, Devlin et al. (2018)) to avoid any forms of bias. Specifically, our input for the masking were the lexical verbs surrounded only by functional elements to retrieve suitable agents and patients. For example, in order to retrieve the arguments for the verb *to break*, we masked the patient leaving the agent in a pronominal (e.g. *she broke (the/a/some/...)* *<MASK>*), and masked the subject of the transitive leaving the patient pronominal to retrieve plausible agents *(e.g. (the/a/some/...)* *<MASK>* *broke it.* The Italian data are built on the English one, with manual intervention when required, to guarantee the acceptability of the sentences, semantic plausibility and a balanced distributions of nouns in terms of gender and number.

Each dataset resulted in 38400 combinations. For the run experiment, we sampled 3000 for each type, following previous publications. Each dataset con-

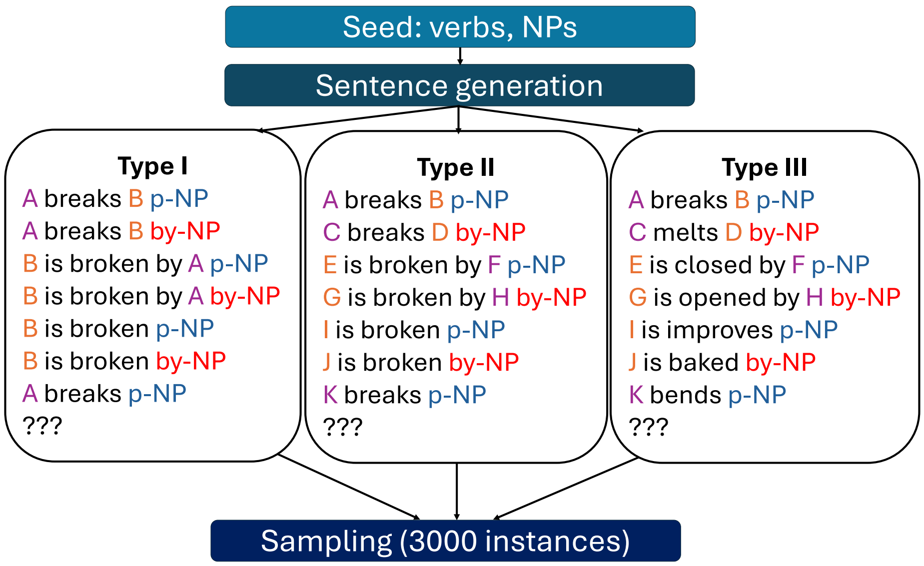

Figure 5: Process generation of the three levels of lexical variation (type I, II, III), exemplified for EnCOS data.

tains 9000 instances of BLMs (3000 for each type), semi-automatically crafted and manually evaluated for plausibility and grammaticality.

3 Baselines

We run a series of experiments based on an architecture proved to show excellent performance in previous works that involve similar tasks (BLMs) or phenomena (verb alternation). In all the experiments, each dataset is split into a 90:10 training/test configuration, allowing for cross-testing across datasets. We evaluate the performance of a baseline across levels (Fun vs. Lex) and lexical variation within the templates (type I, type II, and type III).

We first establish the quality of the performance in the task – quantified by F1 scores – in the same training and same testing configurations (i.e. train on Fun, test on Fun; train on Lex, test on Lex). We then discuss the types of errors found.

We use a multilingual language model and the training data of the model itself vary across English and Italian to make the two languages comparable.[6]

[6]The multilingual dimension can be tested by comparing the performances across languages. It is important to note that the Italian data make a morpho-syntactic difference between the two class of verbs: only the intransitive form of the COS contains an overt (i.e. visible) morpho-syntactic, reflexive-like, marker (e.g. *questa porta **si** apre* 'this door REFLEXIVE opens').

Finally, finer-grained distinctions in terms of learning can be uncovered by exploring the quality, the quantity and the distribution of errors across datasets and their relative training.

3.1 Models

We follow the model and the architectures presented in previous works on BLMs (An et al., 2023; Nastase et al., 2024a,b). As for the sentence embeddings, following previous works on verb alternations (Yi et al., 2022; Samo et al., 2023), we have decided to use the embeddings created by the multilingual model Electra (*google/electra- base-discriminator*). We present the results with a more general architecture, a convolutational neural network (CNN), where the information is regularly distributed in the sentence embeddings which performs overall the worst, and a Variational Autoencoder (VAE) where the relevant information can be distilled from sentence embedding (Nastase et al., 2024a,b).

To avoid data leakage and simultaneously test the abstraction ability with different types of verbs, we use disjoint sets of training and testing verbs. For each experiment on each dataset, we collect the scores of 3 runs.

3.2 Results

Performance FUN and LEX Table 1 shows the F1 scores of all architectures (mean and standard deviation). The overall performance across architectures minimally vary and their difference is not statistically significant (ANOVA; EnCOS: $p = 0.76698$; EnOD: $p = 0.56927$; ItCOS: $p = 0.62741$; ItOD: $p = 0.31152$).

		FUN		LEX	
Dataset	Model	*M*	*SD*	*M*	*SD*
EnCOS	CNN	0.926	0.05	0.836	0.10
	VAE	0.974	0.02	0.919	0.05
EnOD	CNN	0.942	0.05	0.772	0.07
	VAE	0.974	0.03	0.845	0.09
ItCOS	CNN	0.830	0.10	0.536	0.05
	VAE	0.909	0.06	0.584	0.08
ItOD	CNN	0.818	0.10	0.575	0.12
	VAE	0.898	0.05	0.685	0.08

Table 1: Means (*M*) and standard deviation (*SD*) of F1 score across datasets and languages

Figure 6 shows the mean F1 score of the CNN and the VAE for each dataset, based on the variability of lexicalisation (type I vs. type II vs. type III; FUN vs. LEX). The data reveal that testing on functional type I data is, overall, the best-performing configuration. However, differences can be observed across datasets and languages—for example, the COS dataset in English, on which the

BLM is built, generally performs better than the other datasets, except when testing on LEX-type_III-VAE for OD. This difference may be due to the fact that lexical material encodes certain semantic features, such as indicating that the answer is an agent subject (rather than a patient, as in COS) partially in line with previous works with different models in recognizing semantic roles (Proietti et al., 2022).

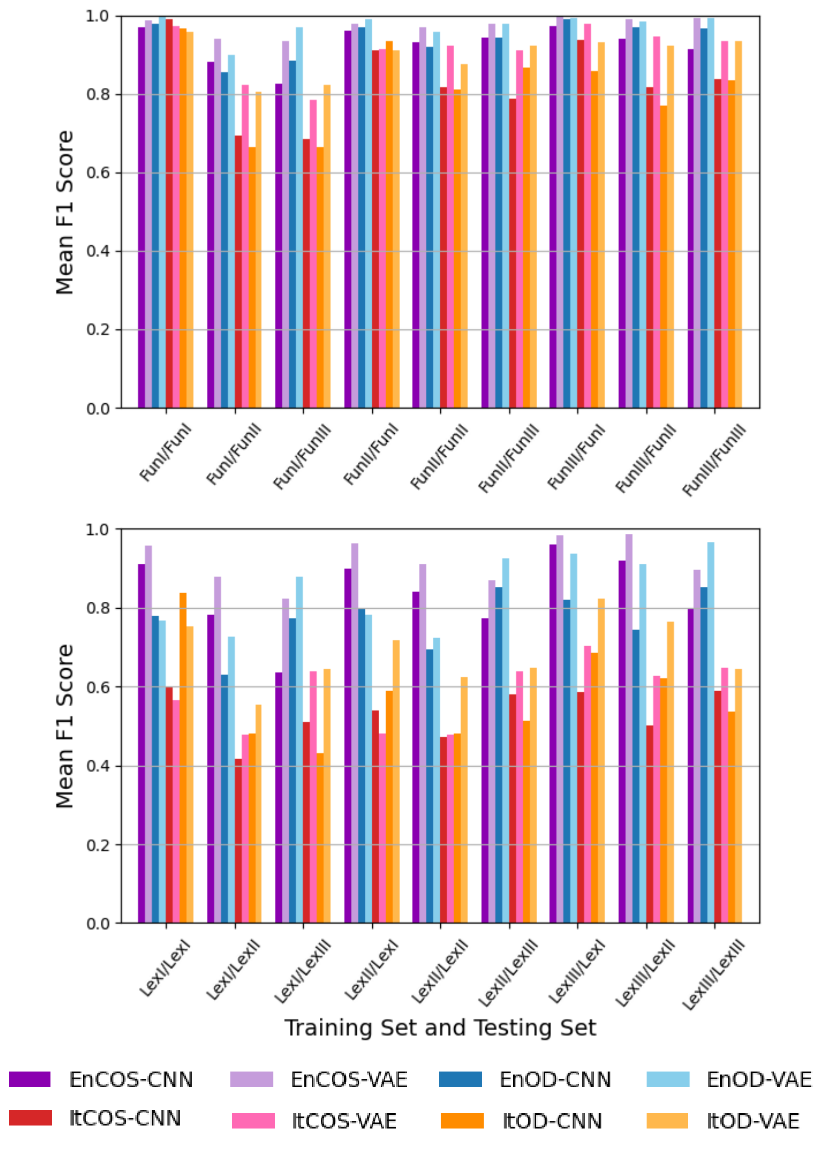

Figure 6: Training and testing across levels of abstractions in Fun (higher panel) and Lex (lower panel) in each dataset and architecture

Cross-testing Whether the model can generalise from one lexical category to the other and see whether abstraction is at work can be observed through the results of cross-testing across classes of datasets (Fun, Lex), as shown in Figure 7. The first notable result is a strong asymmetry with respect to training and testing within the same lexical abstraction. Despite this asymmetry, good F1 scores (greater than 0.60) are observed for EnOD when tested on type I data.

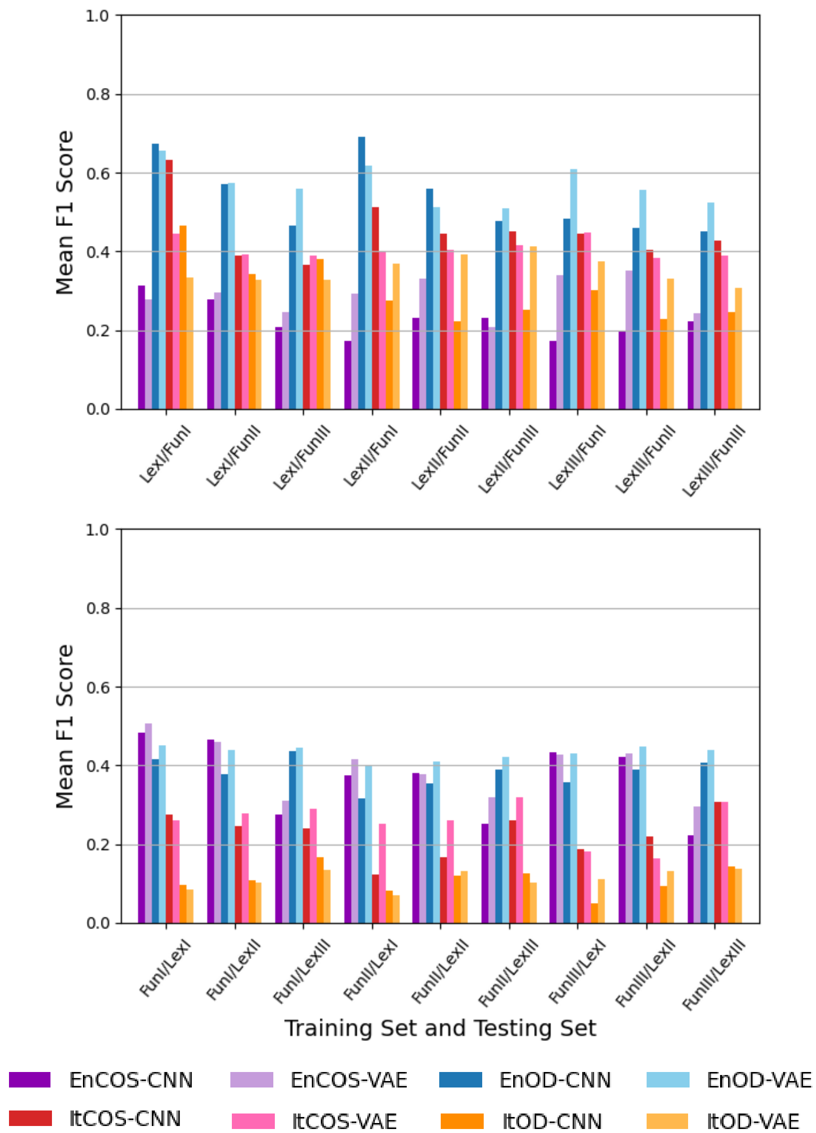

Figure 7: Mean F1 scores - training on LEX, testing on FUN (higher panel) and training on FUN, testing on LEX (lower panel) for each dataset and architecture.

3.3 Error analysis

The error analysis is presented in Figure 8. The most frequent overall mistake is SSM-Int, which is the closest sentence to the target answer, indicating that the model has effectively learnt the syntax of the target answer. This type of error highlights a mistake in the syntax-semantic mapping of the subject; for example, an agent is produced when a patient is expected (e.g., *the clumsy tourist broke* instead of *the vase broke*). The distribution of this type of error, particularly in Italian, shows that it occurs more frequently with COS verbs, where the subject of the active verb is a patient, than in OD contexts, where the subject is an agent. We can interpret this result as indicative of an agent-first bias (see Huber et al. 2023 and references therein). Conversely, the transitive bias (as discussed in Kann et al. 2019 and related work) – detectable by candidate answers labeled as TRANS – is infrequent and can be only found in cross-testing environments.

In this respect, we observe a strong asymmetry between training and testing within the same dataset and between training on one dataset and testing on another. While errors from the same training and testing conditions are very clear in terms of the most prominent error (SSM-Int), we detected a more even distribution of errors during cross-testing in both directions. In this context, EnCOS trainLex-testFun or ItOD trainFUN-testLex exhibit WRBY, which represents an ungrammatical sentence and is observed as a relatively frequent error.

Finally some notes on Italian. The presence of SI in the OD answers, or its absence in the change of state (COS) context in Italian does not represent a frequent source of error. In particular, the results for OD are significant since the presence of SI would result in a transitive structure. This finding indicates that morphosyntactic elements are learned.

An additional control: same vocabulary size for both groups One difference in natural languages between functional and lexical elements is that the former is a closed class. The adopted vocabulary was, in practice, smaller than the lexical ones. To check whether the difference in performance might be due to a bias in the number of items, we created a smaller dataset of lexical elements of similar variability to the functional elements for It-COS (e.g. *il ragazzo ha rotto il borsone, la ragazza ha rotto la borsa, il signore ha migliorato la borsa, etc.*). In the same training/testing configuration, the performance is very high ($M = 0.877$, $SD = 0.08$), However, in the configuration of cross-testing where ItCOS was relatively good (trainLEX test FUN), the results are worse for this limited set of items (train $M = 0.379$, $SD = 0.07$). We can therefore exclude the difference in performance as a by-product of the vocabulary size.

Lingustically-informed datasets The development of this dataset (and similar datasets) requires deep linguistic expertise in the target phenomena. Synthetic, carefully curated data proves essential for investigating linguistic biases in language models, as it is built upon attested exemplars to enable clear observation of patterns and systematic isolation of variables – whether morphological, syntactic, or semantic in nature.

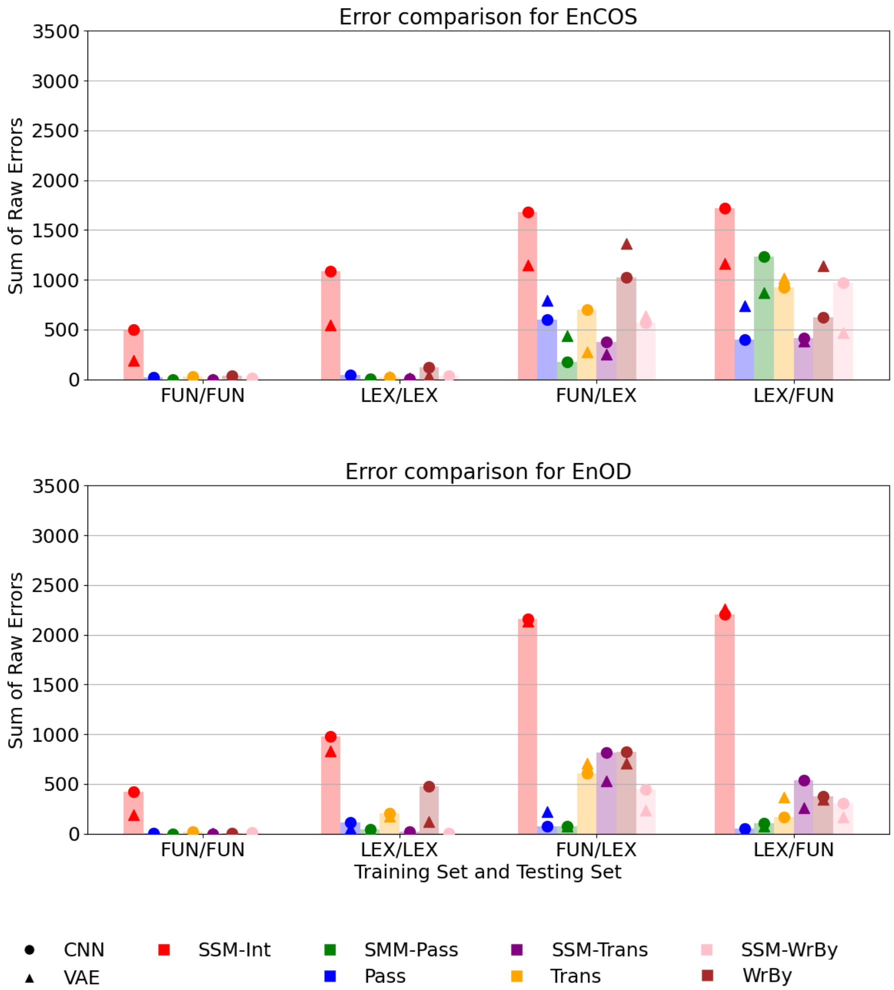

Figure 8: Error analysis across languages, datasets and architectures. The error barchart follows the results of the CNN (circle) intended as a baseline.

The BLM framework provides unique advantages for investigating linguistic phenomena that cannot co-occur in a single sentence. Our methodology suggests how models represent shared structural relationships between alternating constructions that go beyond lexical choices.

Such controlled designs enable precise comparisons between model behavior and established linguistic knowledge. This approach bridges theoretical and computational linguistics by providing rigorous diagnostics for model grammatical knowledge, clearer interpretation of learning mechanisms, and testable hypotheses from the lens of theoretical linguistics (see also Merlo and Samo 2023).

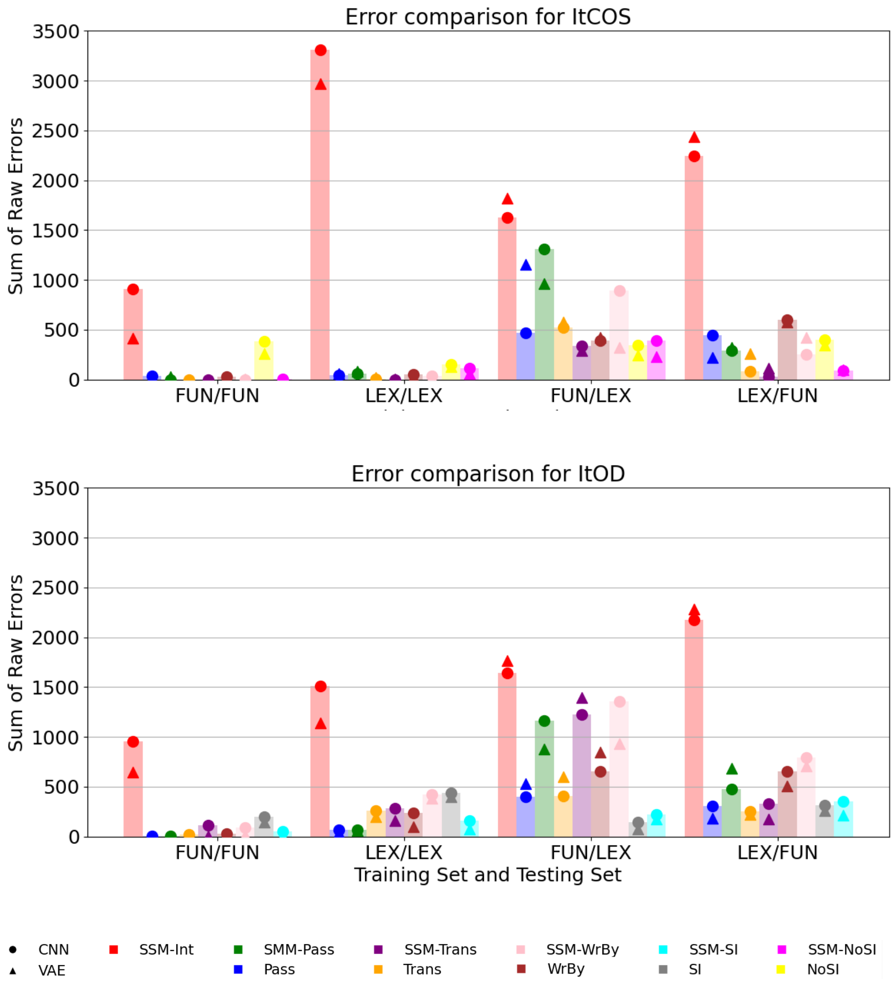

Figure 9: Error analysis across languages, datasets and architectures. The error barchart follows the results of the CNN (circle) intended as a baseline.

4 Conclusions

In this paper, we present BLM-CoS and BLM-OD, structured datasets for English COS and OD alternations in English and Italian following the guidelines of BLMs. This type of data and their explotation might provide interesting results on the understanding of verb argument structure in language models. Future research should expand the languages as well the verb classes.

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